

CRISTOFF STAN

Senior Software Developer

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PROFESSIONAL SUMMARY

Experienced Senior Software Developer with a strong background in modern C++ (C++20), 3D graphics, and performance-critical desktop applications. Proven ability to design and optimize high-performance tooling for simulation, CAD environments, and digital modeling workflows. Collaborative, quality-driven, and focused on delivering user-centric features that balance speed, reliability, and usability.

TECHNICAL SKILLS

- **Languages:** C++20, Python, JavaScript, TypeScript, SQL, C#
- **Frameworks/Libraries:** Qt 5/6, OpenGL, Vulkan (basic), Boost, STL, Eigen
- **3D Technologies:** 3D Math (vector/matrix algebra, quaternions), Scene graphs, Simulation systems
- **Tools:** Visual Studio, CMake, Git, Conan, Perforce, Jira, Unity Editor (light)
- **Performance/Optimization:** Multithreading, Memory management, Profiling (Valgrind, VTune), SIMD
- **Others:** CAD principles, Prefab systems, UI/UX architecture, Serialization formats (JSON, XML, Protobuf)

WORK EXPERIENCE

Sensidev

Senior Software Engineer

September 2021 – December 2024

Timișoara, Romania

- Spearheaded the development of a performance-optimized C++ module for simulating mechanical part interactions in a 3D modeling tool, reducing lag by 38% in large assemblies.
- Created extensible UI tooling using Qt for interactive manipulation of scene elements, ensuring responsiveness even with complex object hierarchies and nested prefabs.
- Engineered serialization logic for project files using a hybrid JSON + binary approach, enabling faster load times and compatibility with existing CAD pipelines.
- Collaborated with product managers and industrial designers to prototype advanced design assist tools, including smart snapping, measurement overlays, and geometry validation.
- Designed and implemented a dynamic scene graph system that efficiently manages spatial hierarchies and supports real-time simulation playback and rollback.
- Integrated a GPU-based acceleration path using OpenGL compute shaders to improve rendering throughput for large meshes and texture-heavy assets.
- Devised a testing and debugging suite for geometry-related bugs, including invalid normals, T-junctions, and overlapping faces, which cut issue triage time by 60%.
- Led code reviews, mentored junior engineers, and introduced modern C++20 best practices, enhancing code safety and maintainability across modules.

Syscom Digital
Software Engineer
October 2019 – August 2021
Bucharest, Romania

- Developed modular, reusable C++ components integrated with Qt Designer for fast prototyping of graphical interfaces used in embedded simulators.
- Built a set of tools for visual debugging of 3D camera angles and mesh alignment, supporting marketing visualization and product showcase scenes.
- Contributed to a simulation subsystem that replicated physical interactions between components, using deterministic update loops and interpolation.
- Enhanced legacy OpenGL-based rendering paths by implementing view frustum culling and geometry batching, improving frame rates by over 45%.
- Led the migration from outdated custom serialization format to a more robust schema using Protobuf and XML for CAD data interchange.
- Conducted memory leak and race condition audits using Valgrind and TSAN, eliminating multiple hidden bugs in multithreaded rendering routines.
- Participated in weekly sprint planning and roadmap discussions with designers and stakeholders to prioritize simulation tool feature delivery.
- Worked closely with QA teams to define automation test cases for CAD-specific scenarios including cross-section analysis and dimension constraints.

Soft Build
Software Developer
June 2016 – July 2019
Timișoara, Romania

- Developed desktop engineering tools in C++ using Qt and OpenGL, supporting architectural modeling and thermal simulation workflows.
- Implemented an interactive grid snapping and orthogonal constraint system for 2D sketching tools within a larger 3D environment.
- Designed memory-efficient data structures for mesh topology storage, reducing RAM usage by 27% in large blueprint projects.
- Created a plugin system allowing external teams to extend modeling functionalities without modifying the core application.
- Tuned rendering pipelines to support real-time feedback during model editing, introducing incremental mesh updates instead of full re-renders.
- Conducted in-depth performance profiling to optimize data access patterns and reduce cache misses during complex simulations.
- Collaborated with academic partners on CAD tool integrations that supported experimental design workflows for engineering students.
- Introduced versioned project saving features with backward compatibility, facilitating easier long-term project evolution and support.

Decima Digital
Software Developer
November 2015 – April 2016
Tallinn, Estonia

- Built internal visualization tools using Qt and OpenGL to assist in previewing architectural renderings for urban planning clients.
- Implemented a custom scene composition editor with layered canvas and object snapping features for drag-and-drop operations.

- Used C++ STL and multithreading to generate preview thumbnails for hundreds of models in parallel, dramatically cutting processing times.
- Added undo/redo logic for complex modeling operations using command pattern with stack-based history tracking.
- Defined serialization protocols for asset state storage in JSON format to ease external asset import/export tasks.
- Participated in UI/UX feedback cycles with clients to improve editor usability and onboarding for new users.
- Documented internal 3D math libraries for reuse across teams and provided usage examples to reduce duplication of effort.
- Integrated a custom licensing system based on encrypted token validation, ensuring proper license tracking for enterprise clients.

Borna Technology

Software Developer Intern

September 2014 – August 2015

Tallinn, Estonia

- Contributed to the development of real-time visualization widgets for monitoring IoT sensor data using C++ and Qt.
- Supported migration of legacy 2D simulation tool to a hybrid 2D/3D visualization platform using OpenGL overlays.
- Assisted in implementing vector math and transformation utilities used in coordinate system mapping.
- Performed unit and integration testing of core modules and implemented test harnesses for geometric calculation functions.
- Created internal tools to batch-process 3D assets for automated format conversion and normalization.
- Helped develop cross-platform build scripts using CMake and shell scripting for Linux and Windows targets.
- Reviewed and documented simulation flows and proposed modularization strategies for better code reuse.
- Participated in intern mentoring program, giving peer code walkthroughs and contributing to shared internal wiki.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,

2010 – 2014

CERTIFICATIONS

- ❖ Advanced C++ Programming – Coursera / University of Illinois
 - ❖ 3D Graphics Programming with C++ and OpenGL – Udemy
 - ❖ Qt 6 UI Development for Desktop Applications – Udemy
 - ❖ Data Structures and Algorithms in C++ – Pluralsight
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PROJECTS

✓ **CADXSim** – Internal Simulation Toolkit for 3D Model Validation

Built core simulation loop and real-time constraint solver for a CAD modeling product, enabling rapid prototyping and stress testing of virtual assemblies.

✓ **GeoScene Editor** – Qt-based 3D scene editor

Developed custom scene editing interface with drag-drop hierarchy, transform gizmos, and prefab support. Used in architectural design simulation workflows.